LESSON PLAN

NDT (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Ultra-sonic inspection, basic principle waves propagation characteristics acoustic impedance basic inspection methods, ultrasonic equipment
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Ultra-sonic inspection, basic principle waves propagation characteristics acoustic impedance basic inspection methods, ultrasonic equipment, data representation, transducer, various probe couplant
5. Specific Objectives	19.1.4.1 Trainee will be able to describe the method of ultra-sonic inspection.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Ultrasonic Equipment setup, coupling agent application, and data representation on screen under instructor supervision.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary frequency range used in ultrasonic inspection?
(b) Define acoustic impedance in the context of wave propagation.
(c) List two common couplants used in NDT applications.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic principle of ultrasonic inspection • Waves propagation characteristics • Acoustic impedance definition and effect	L & Ds	PPT, IB	15 min
(b)	• Ultrasonic equipment components • Transducer types and functions • Various probe configurations	L & Ds	PPT, PDF	20 min
(c)	• Basic inspection methods overview • Data representation techniques • Couplant selection and application	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how acoustic impedance mismatch affects wave reflection.
(b) Describe the role of the couplant in ultrasonic inspection.
(c) Differentiate between pulse-echo and through-transmission methods.

12. Home Assignment and Follow Up	Prepare a brief note on the advantages and disadvantages of ultrasonic inspection compared to optical holography (referencing LP 08 syllabus).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Optical holography basic principle, procedure for setting up construction and re-construction, laser, interferogram advantages disadvantages and application	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min